DSGE Model Estimation via VAR IRF Matching: The Short and Long of It

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Motivation

- VARs are often used to estimate DSGE frameworks by IRFs minimization matching
- IRFME $\hat{\zeta} = \arg\min_{\zeta} (\hat{\Psi} - \Psi(\zeta))' V^{-1} (\hat{\Psi} - \Psi(\zeta))$

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Ramey (2016) p. 83

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  Ramey (2016) p. 83
- Misspecification can deliver distorted results mis informing the central banker

This Paper

Research Question:

- Can VAR **truncation bias** have significant implications for DSGE estimation?

What it's done:

- Revisit ACEL RoED (2011) contribution by estimating Homogeneous vs Firm Specific versions of their DSGE with VARs with different dynamic structures

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Findings:

- Increasing the VAR lag structure cast doubds on the importance of Variable Capital Utilization to explain inflation inertia
- Lag choice can have implicationtions for optimal policy desing

VAR IRFs DSGE Parameters

Roadmap

- Related Literature
- ACEL model
- DSGE as VAR(∞)
- Truncation Bias
- Results

Related Literature

Long-Lag VARs:

- De Graeve and Westermark, (2025); Baumeister, (2025); Antolín-Díaz and Surico, (2025); Antolín-Díaz et al., (2026)

Truncation Bias:

- Ravenna, (2007); De Graeve and Westermark, (2025); Chari et al., (2008); Barnichon et al., (2026)

DSGE State Space representation into VAR(∞)

- Fernández-Villaverde et al., (2005); Ravenna, (2007); Giacomini, (2013); Theodoridis, (2011); Forni et al., (2016)

DSGE IRFs matching with VARs

- Rotemberg and Woodford (1997); Christiano et al., (2005); Boivin and Giannoni (2006); Altig et al., (2011); Rossi et al., (2012)

CEE and ACEL's DSGE Motivation

- Standard NK models replicate inflation persistence with levels of price stikiness, inconsistent with micro data evidence
- CEE JPE (2005) and ACEL RoED (2011) seminal works introduced the concept of **firm specific capital** to model inflation persistence, mechanism supported by IRF matching with VAR(4)

ACEL's DSGE Mechanism

- In Altig et al., (2005) inflation is inertial despite firms re-optimising prices on average every 1.5 quarters
- Inertia is explained by the fact that, when firms re-optimise, they do it by a small amount because each firm's short run marginal cost is increasing in its own output
- The key assumption is that, in any given period, a firm's capital stock is pre-determined and firm-specific

They build two models on Christiano et al. JPE, (2005):

- Homogeneous capital
- Firm-specific capital

ACEL's DSGE Mechanism (Cont'd)

- When log-linearized, the only difference in the characterizing equilibrium equations lies in the Phillips Curve

$$\Delta\hat{\pi}_t = \mathbb{E}[\beta\Delta\hat{\pi}_{t+1} + \gamma\hat{s}_t|\Omega_t] \tag{1}$$

$$\gamma = \frac{(1-\xi_p)(1-\beta\xi_p)}{\xi_p}\chi \tag{2}$$

- $\hat{s}_t$ is the economy-wide marginal cost, γ the Phillips Curve slope, ξ_p the Calvo price stickiness parameter
- χ reduces the sensitivity of inflation to marginal costs in the firm-specific capital model
- The two models are observationally equivalent in terms of prices and quantities: macro data identify γ, so the difference between the two models lies in the implied ξ_p

ACEL's DSGE Mechanism (Cont'd)

Here χ is a non-linear function representing the strategic complementarity in price settings

- In the **Homogeneous** model, $\chi = 1$: γ depends only on β
- In the **Firm-specific** model, $\chi < 1$ and captures two forces that make a firm move its price *less* when it can reset:
  - It depends on Variable Capital Utilization σ_a
  - The costlier it is to vary capital utilization, the steeper the firm's own marginal cost curve
  - The more elastic the firm's demand, the larger the fall in output (and hence in marginal cost) after a price increase

- So χ governs **how much** a firm moves its price
- ξ_p **how often**, with implied duration $1/(1-\xi_p)$

Since γ is estimated by aggregate data, a smaller χ imply a smaller ξ_p

ACEL (2005) Results

The estimated γ is 0.040, which implies:

- **Homogeneous** capital: $\xi_p = 0.82$, firms change prices every 5.6 quarters
- **Firm-specific** capital: firms change prices every 1.5 quarters

The gap arises because firms' demand curves are highly elastic and their marginal cost curves are very steep: the same γ is delivered by frequent but small price adjustments.

Only the firm-specific capital model reconciles the micro-macro pricing conflict

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- **Are these results robust to an enriched lag structure of the VAR?**

DSGE, State Space and VAR(∞) Representation

- Many DSGE's log linearized solution can be mapped in State Space representation and then into VAR(∞)
- When forecasting information criteria are not used
- Standard lag choice by SVAR researchers is 4 for quarterly data and 12 for monthly
- If a DSGE is estimated via IRFME with a finite order VAR, the DSGE is estimate with a Biased by Truncation VAR
- VARs are typically small systems with few lags often interpret as models of the entire economy, IRFs are highly sensitive to variable and lag choice

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- **Should the VAR lag order be increased to overcome Truncation Bias implications?**

Illustration of the problem: Truncation Bias

$$Y_t = B_1 Y_{t-1} + \cdots + B_p Y_{t-p} + u_t, \qquad \mathbb{E}(u_t u_t') = \Sigma \tag{3}$$

$$B(L) = B_1 L + \cdots + B_p L^p \tag{4}$$

- The VAR with $p < p^*$ suffers from truncation bias.[1]
- Lag truncation biases the reduced form coefficients $B(L)$ and the reduced form variance covariance matrix Σ, which are the two main ingredients for IRF estimation.

$$\text{IRF} = f\big(B(L), \Sigma\big) \tag{5}$$

[1] Where p^* denotes the DGP true lag length.

Long-Lag VARs

De Graeve and Westermark, (2025) demontrates that increasing the lag structure improves VAR's IRFs estimates obtaining:

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Lower Bias

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Lower Bias

- By reducing the truncation bias misspecification at each horizon

Lower Variance

- Misspecification reduction lowers the variance of the residuals
- This variance reduction counteracts the imprecision caused by increased parametrisation

Long-Lag VARs Benefits

The variance-reducing effect of lower misspecification often dominates the penalty of increased parameterisation.

Compared to short-lag VARs, Long-Lag VARs have:

- Reduced truncation bias
- More precise inference
- Lower MSE
- Better coverage

Residuals' Variance Reduction

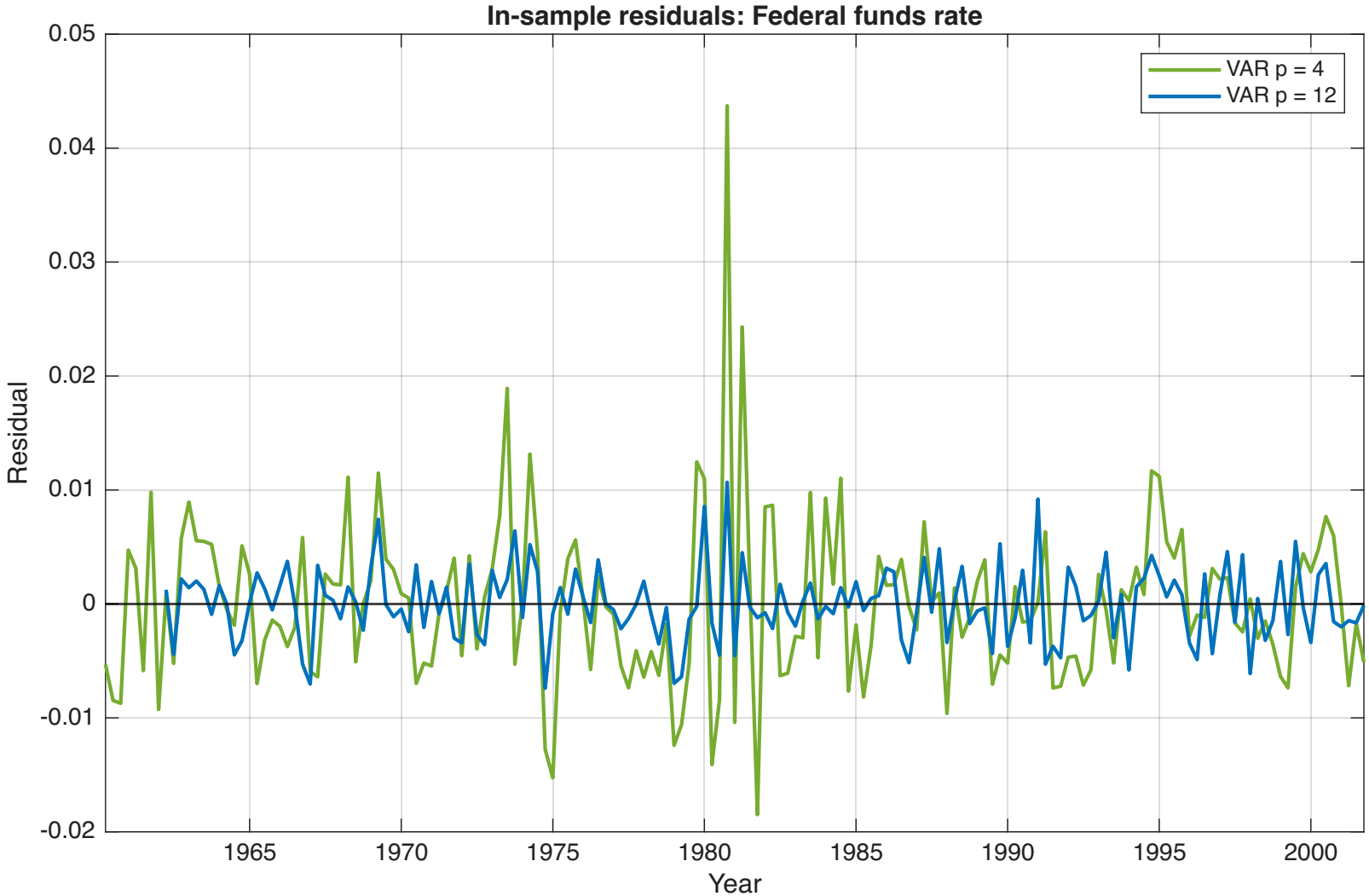


Figure 1: FFR Residuals Comparison Across lag 4 Vs 12

Portmanteau P-Values Across lags

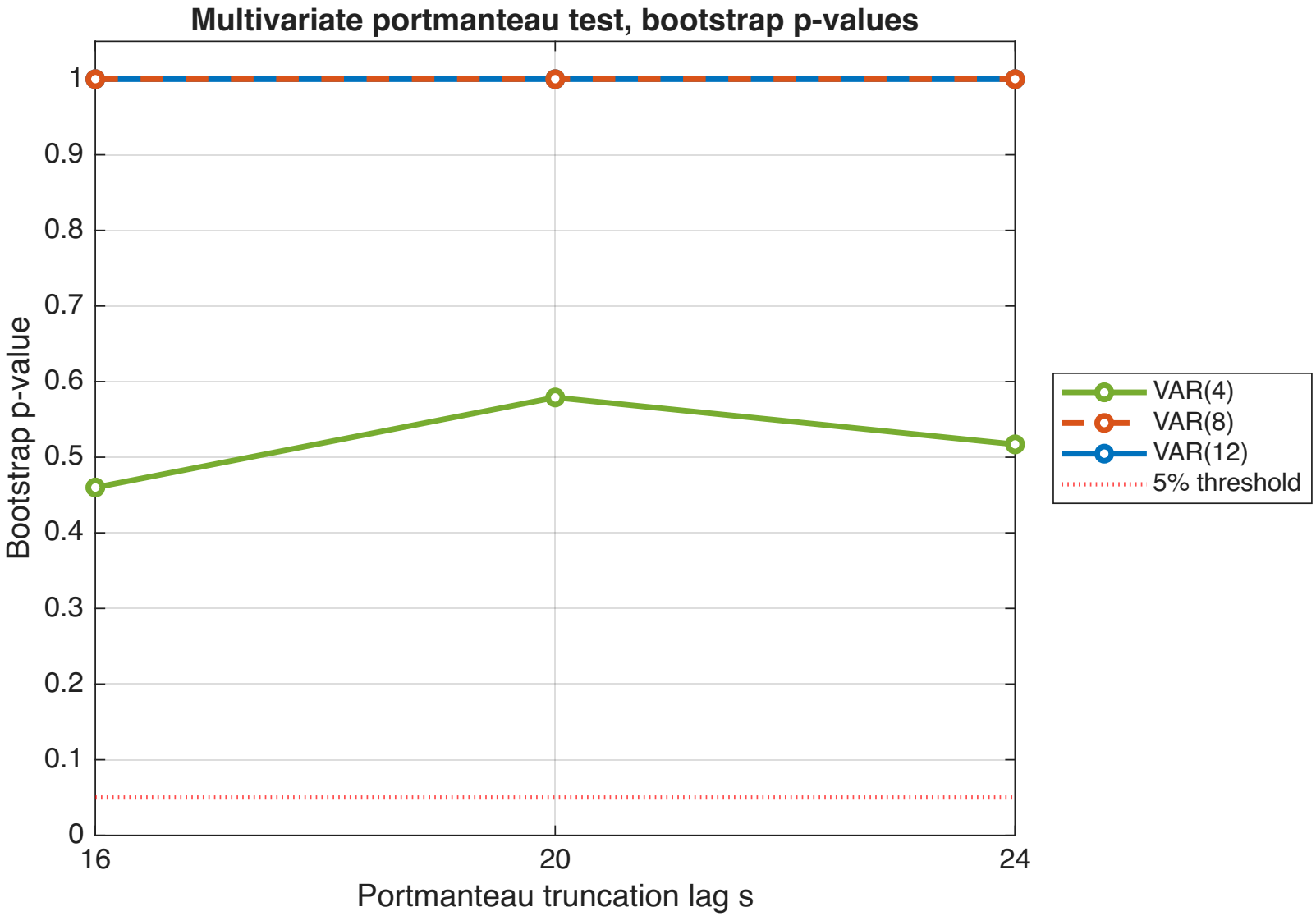


Figure 2: Portmanteau P-Values Across lags

Replication Exercise

Framework:

- **Altig et al., (2011)**'s DSGE[2]

VAR:

- Short and Long-Run restrictions, medium scale 10 variables
- Monetary Policy, Neutral and Embodied Technology shocks
- Sample period 1959:Q2-2001:Q4

Exercise:

- I re-estimate the VAR increasing lag order
- Re-estimate the DSGE by IRFs matching
- **Analyse implications of possible variations in the structural parameters and IRFs across different lag choice**

[2]Where the replication package provided by the authors refers to the 2005 NBER working paper

VAR IRFs; Monetary Policy Shock

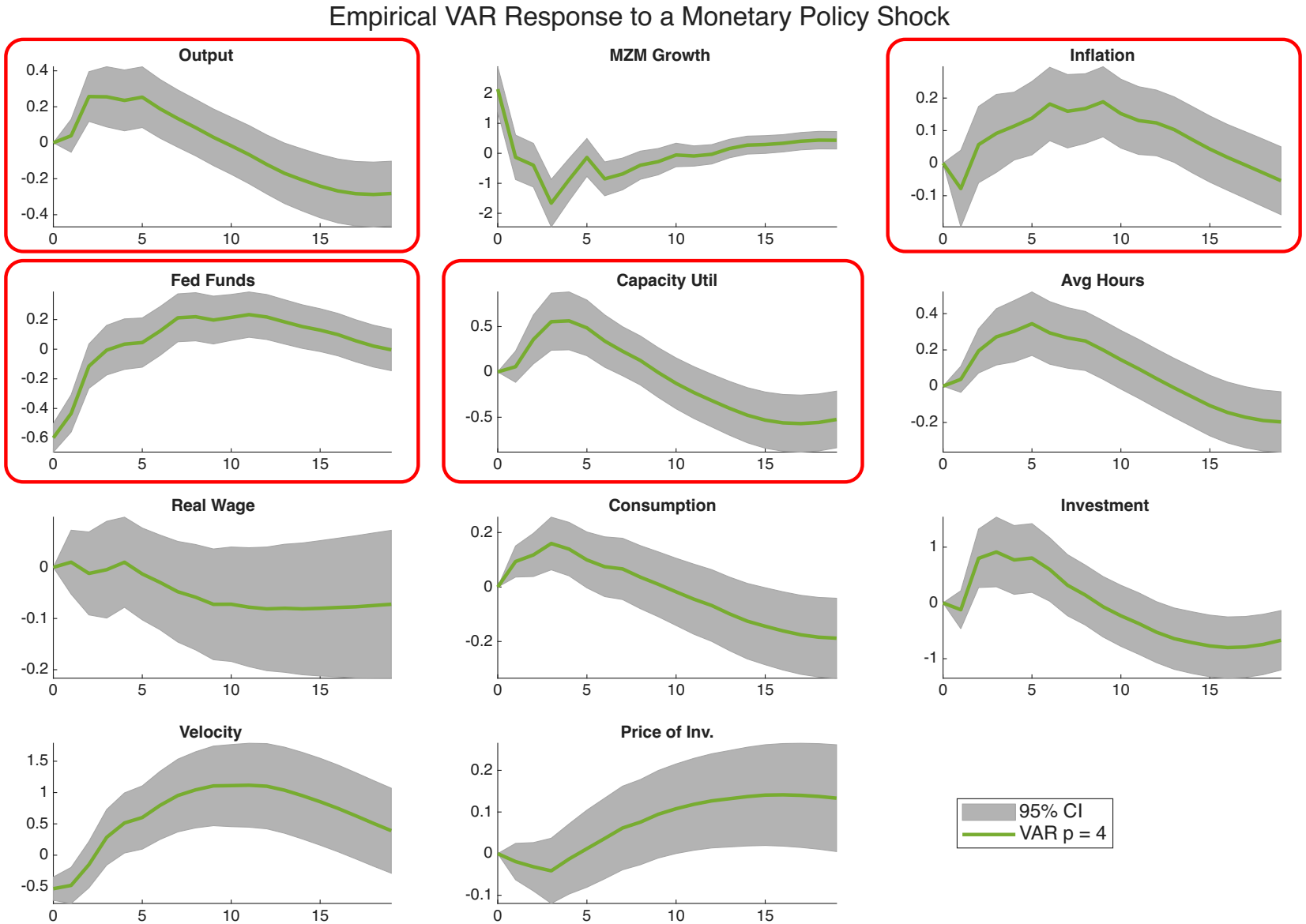


Figure 3: VAR IRFs Comparison: Monetary Policy Shock: 4 lags

VAR IRFs; Monetary Policy Shock

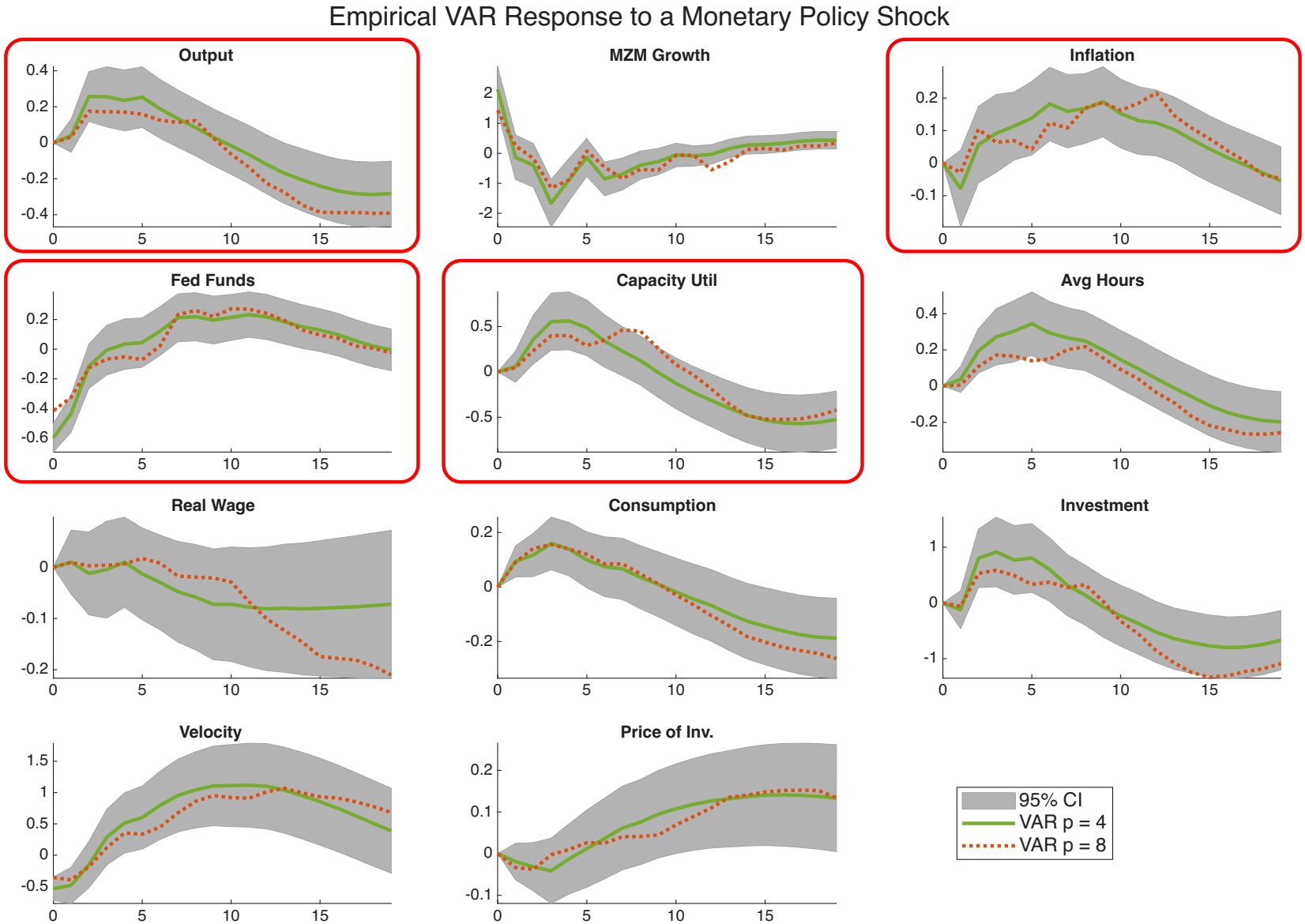


Figure 3: VAR IRFs Comparison: Monetary Policy Shock: 8 lags

VAR IRFs; Monetary Policy Shock

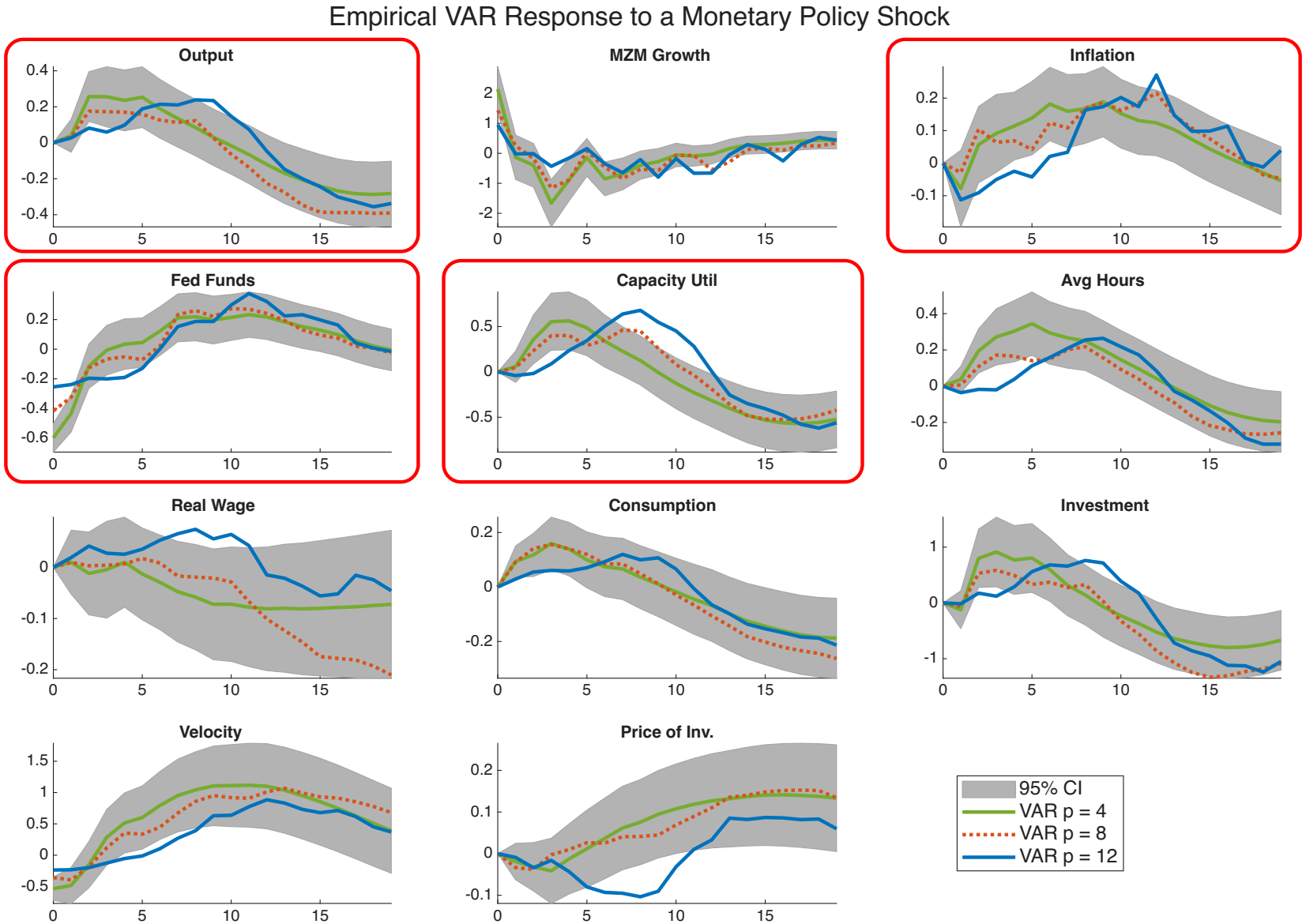


Figure 3: VAR IRFs Comparison: Monetary Policy Shock: 12 lags

What about DSGE's structural parameters?

DSGE Structural Parameters

Parameter	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Structural Estimated Frictions			
Price markup (λ_f)	1.0100	1.0100	1.0100
	(-)	(-)	(-)
Wage stickiness (ξ_w)	0.7221***	0.5617***	0.4838***
	(0.1506)	(0.1142)	(0.0757)
Phillips curve slope (γ)	0.0402	0.0402	0.0129***
	(0.0246)	(0.0598)	(0.0042)
Capacity util. cost (σ_a)	1.9938	0.1175	0.0042
	(2.6066)	(0.6299)	(0.0230)
Habit formation (b)	0.7066***	0.6935***	0.8596***
	(0.0463)	(0.0570)	(0.0203)
Inv. adj. cost (S'')	3.2807*	3.0372*	6.8431***
	(1.8874)	(1.5506)	(2.3306)
Money demand elast. (ϵ)	0.8077***	0.7531***	0.8346***
	(0.2111)	(0.1473)	(0.1944)
Implied Price Stickiness			
Homogeneous capital ($\xi_{p,hom}$)	0.8213***	0.8213***	0.8958***
	(0.0480)	(0.0574)	(0.0139)
Firm-specific capital ($\xi_{p,fs}$)	0.3372**	0.6448***	0.8884***
	(0.1505)	(0.1285)	(0.0220)

Note: Standard errors in parentheses. The Calvo parameters are recovered numerically from the estimated structural parameters; their standard errors come from a 1,000,000-draw Monte Carlo of the joint distribution of the estimated structural parameters.

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Implied Duration

Model Specification	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Homogeneous Capital Model			
Calvo parameter ($\xi_{p,hom}$)	0.82	0.82	0.90
Implied duration (quarters)	**5.6**	**5.6**	**9.6**
Firm-Specific Capital Model			
Calvo parameter ($\xi_{p,fs}$)	0.34	0.64	0.89
Implied duration (quarters)	**1.5**	**2.8**	**9.0**

Note: Implied duration is calculated as $D = 1/(1 - \xi_p)$.

- Firm Specific model's Calvo Price parameter converges to the Homogeneous one
- Back to the micro-macro implied price duration missmatch

Conclusions

Following the advices from the Long-Lag VARs literature. Replicating ACEL et. al., (2005), enriching the VAR lag structure I find that:

- Capacity Utilization Cost, crucial variable to justify the firm specific model, dramatically shrinks to 0
- Firm specificity of capital seems not the key concept to explain inflation inertia
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De Graeve & Westermark (2025) and Baumeister (2025)

- Long-Lag VARs migth have structural implication, it's up to the researcher to demonstrate that its findings are robust to an encriched lag structure

Thanks a lot for your attention!

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VAR IRFs All CI; Monetary Policy Shock

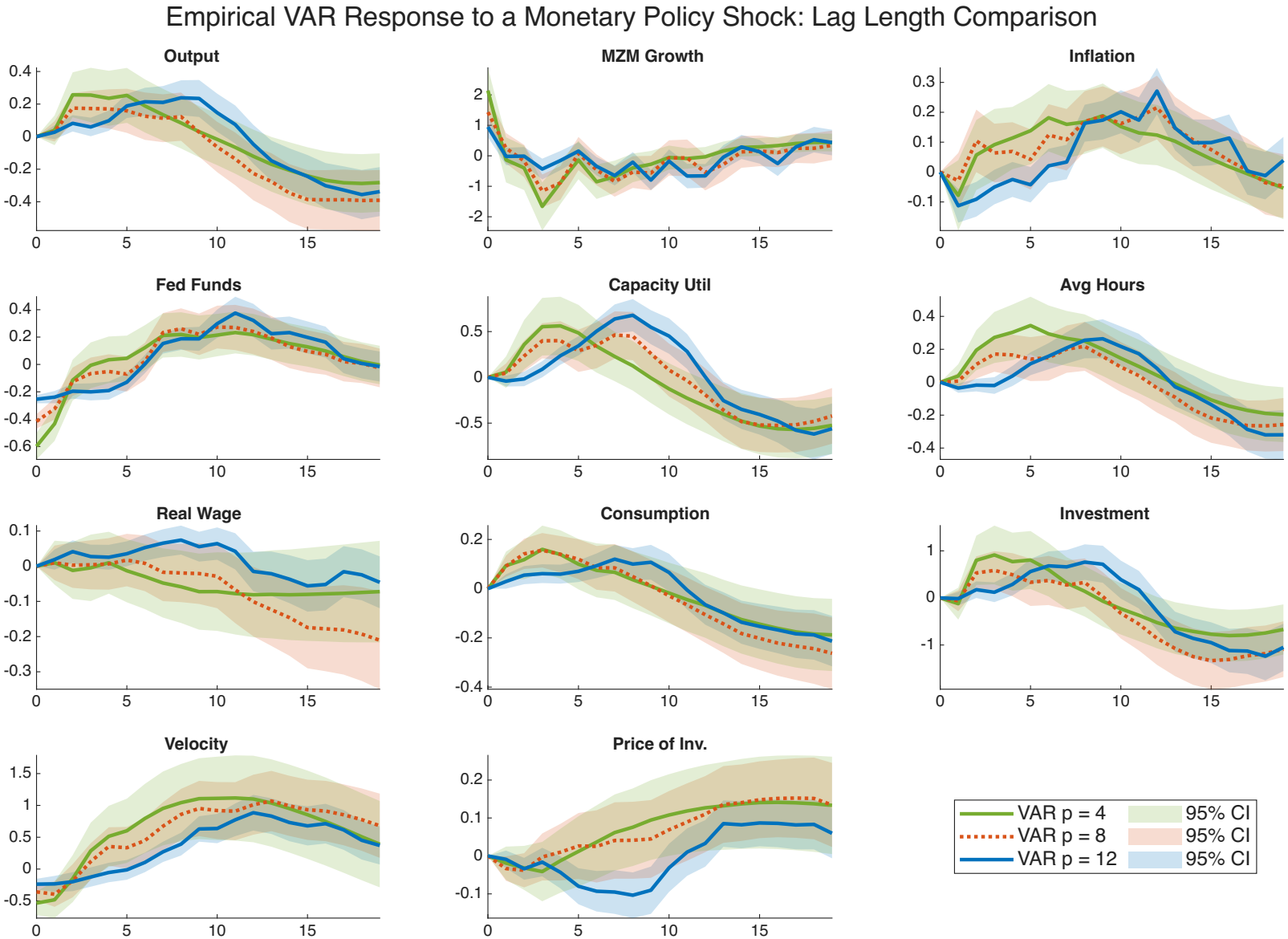


Figure 4: IRFs VARs all CI Comparison: Monetary Policy Shock

Bootstrap Confidence Intervals Shrinks!

- With respect to De Graeve and Westermark (2025) examples, this exercise have a medium scale VAR; 10 variables + constant
- Estimating the VAR 12 we reach the feasible limit of the degrees of freedom T-k = 30 and total numebr of coefficients $n + n^2 p = 1210$
- Bootstrap Confidence Intervals actually shrinks!
- Confirming De Graeve and Westermark (2025) p.2 JME point on variance reduction even in a limit case "Misspecification reduction not only leads to Bias-Variance trade off, but it reduces variance"

DSGE IRFs; Monetary Policy Shock

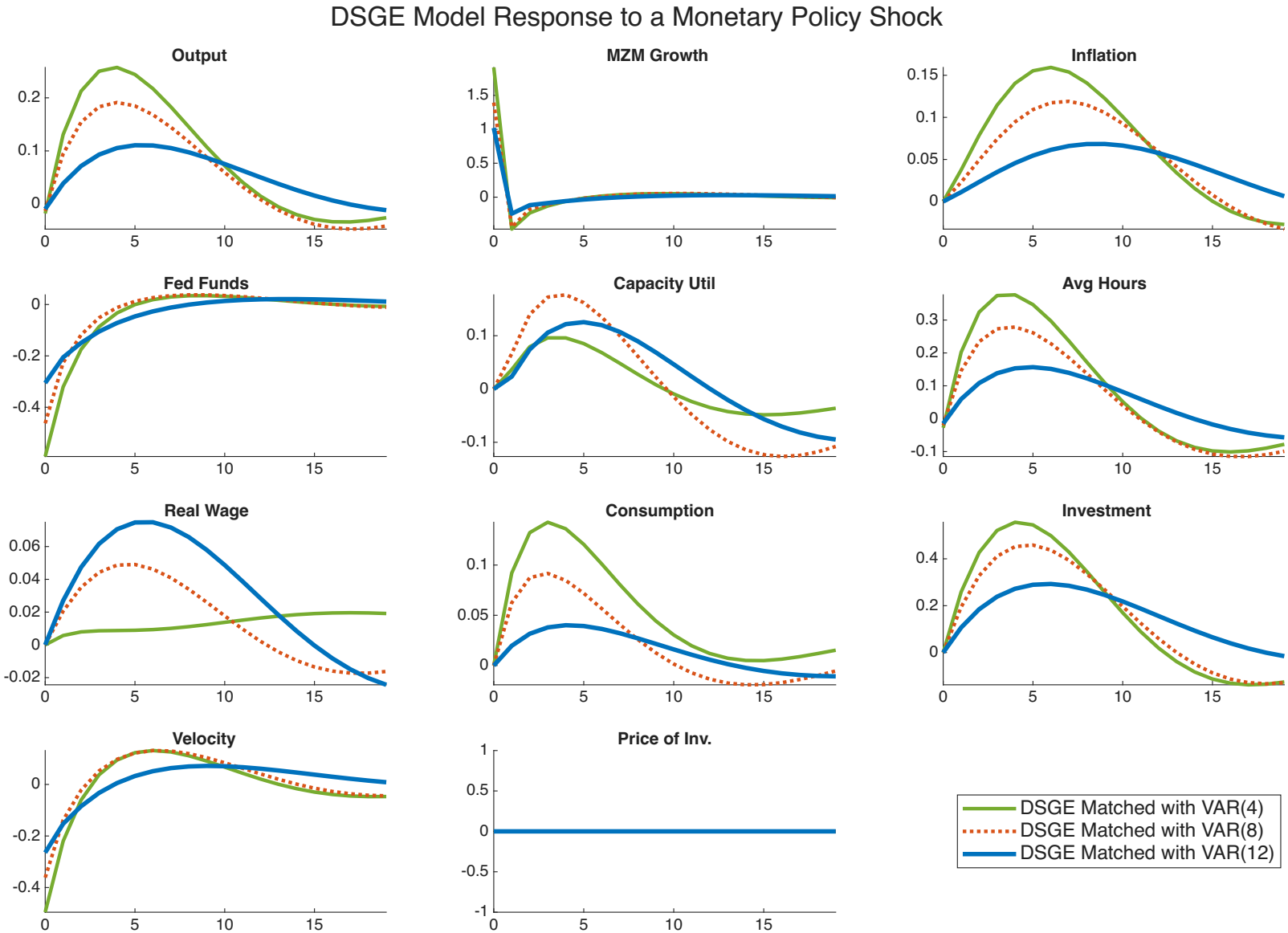


Figure 5: DSGE IRFs Long Lag Progressively Plots 4 8 12

VAR IRFs; Neutral Technology Shock

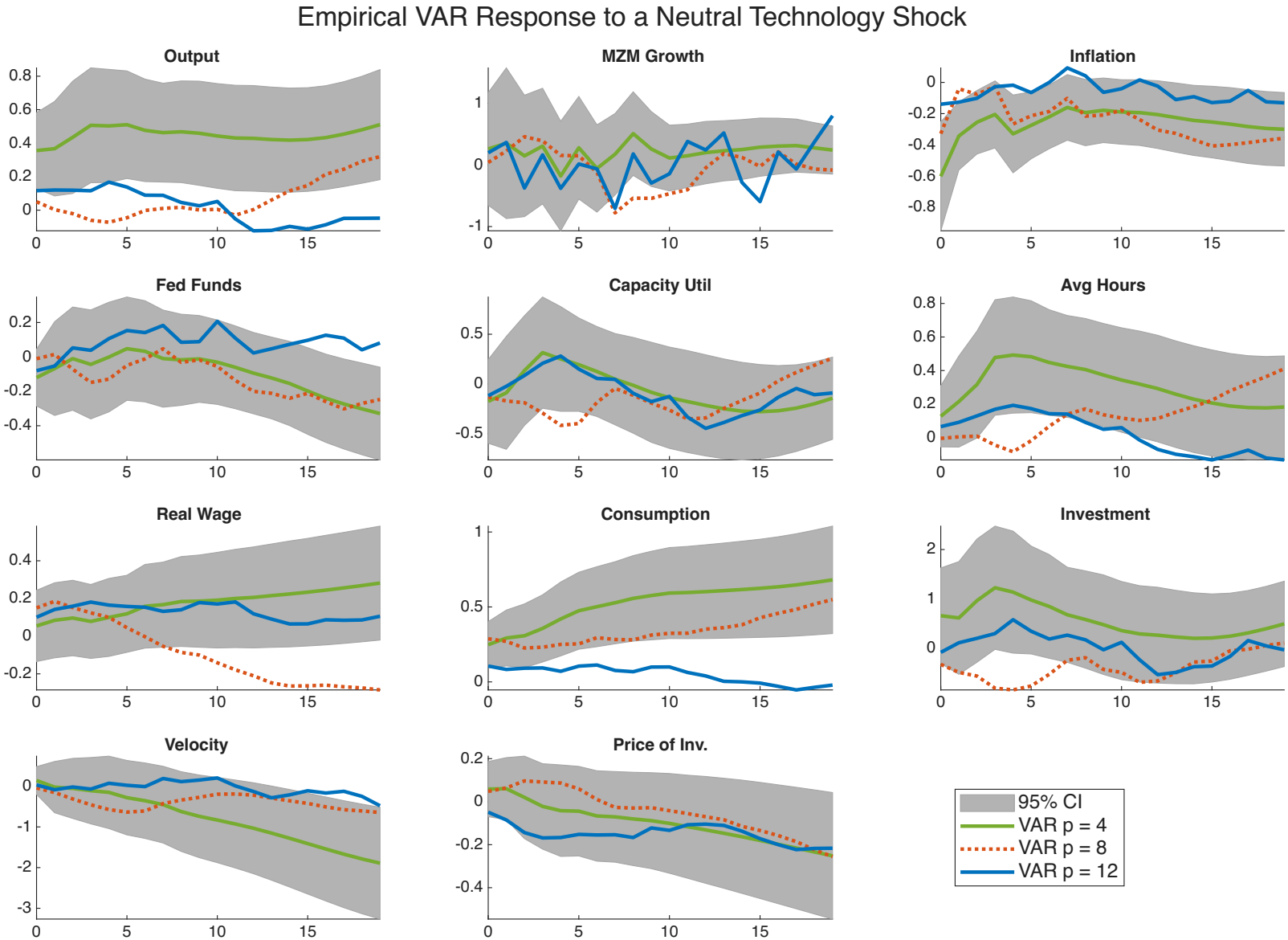


Figure 6: IRFs VARs all CI Comparison: Neutral Technology Shock

VAR IRFs; Embodied Technology Shock

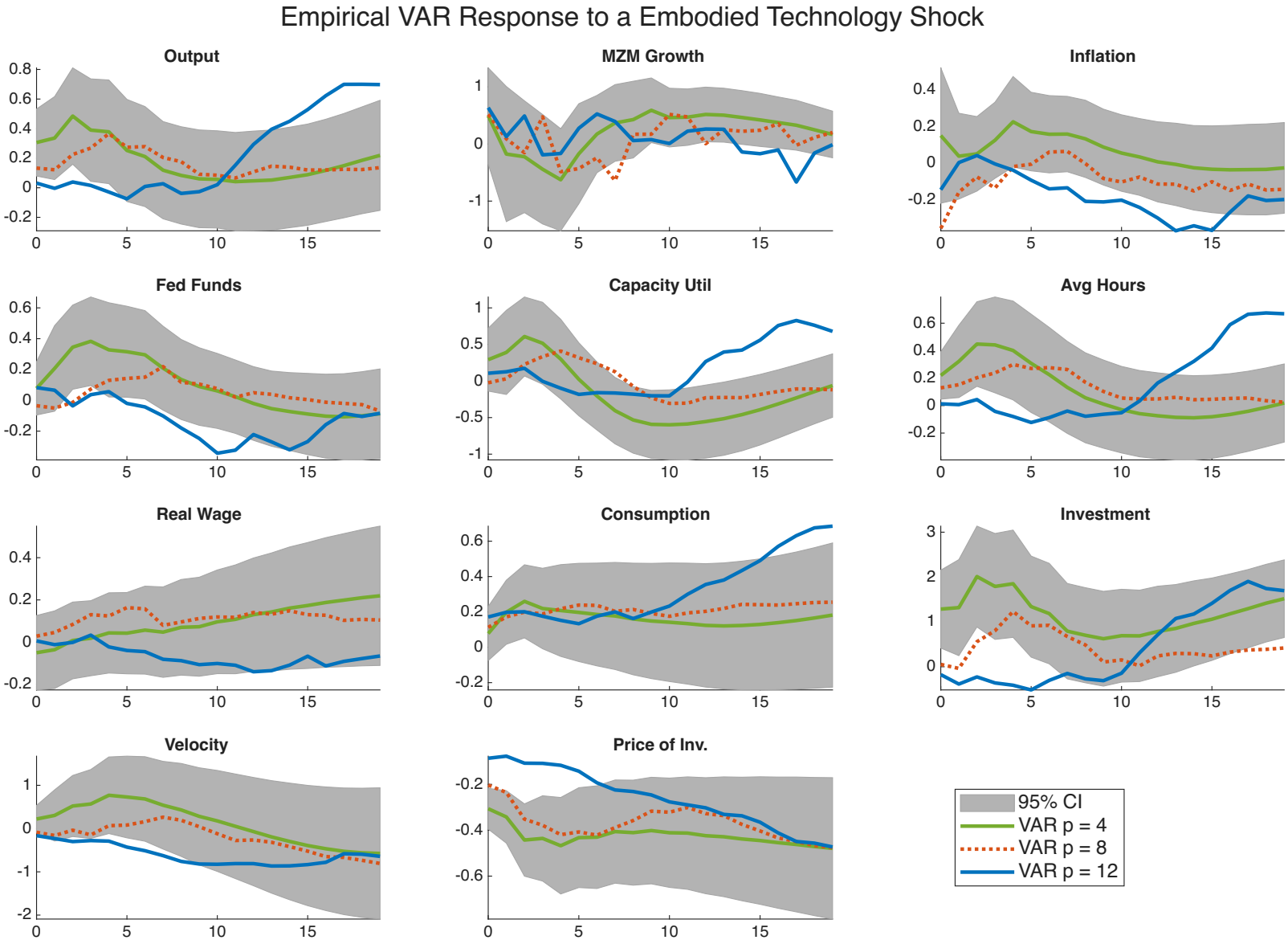


Figure 7: IRFs VARs all CI Comparison: Embodied Technology Shock

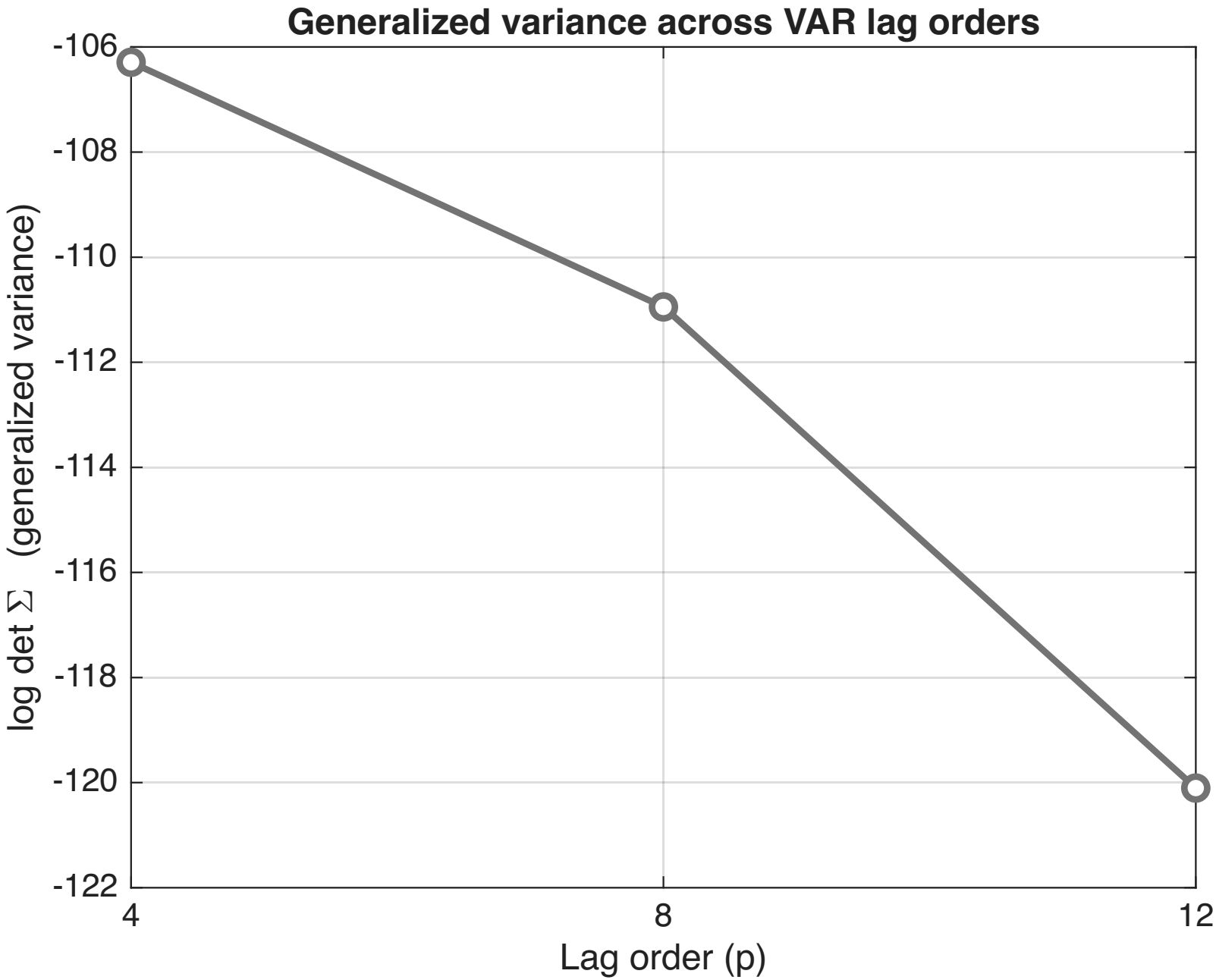
RMSE Table

Table 1: In-sample RMSE of the reduced form VAR residuals

Variable	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Δ ln rel. price of investment	0.3387	0.2752	0.1962
Δ ln GDP / hours	0.5220	0.3758	0.2470
Δ ln GDP deflator	0.2095	0.1823	0.1190
Capacity utilisation	1.1726	0.8427	0.5110
ln hours	0.4977	0.4032	0.2410
ln(GDP/hours) - ln real wage	0.5469	0.4629	0.3278
ln consumption / GDP	0.5690	0.4374	0.3059
ln investment / GDP	1.8667	1.4188	0.9546
Federal funds rate	0.7179	0.5340	0.3201
Velocity: ln P + ln GDP - ln MZM	1.4146	1.0896	0.7643

- RMSE of the RF residuals $u_t = A_0^{-1}\varepsilon_t$, in percentage points
- Lag orders evaluated on its own sample (T = 167, 163, 159)
- Restricting every lag order to the common last 159 observations changes no entry by more than 0.0927 pp.

Generalized Variance



Generalized Variance

Table 2: Generalized variance across VAR lag orders

Lag order	$\log\det\hat{\Sigma}$	Δ **vs.** $p=4$	Δ **Variance (%)**	Δ **St. Dev. (%)**
$p=4$	-106.2903	0.0000	+0.0%	+0.0%
$p=8$	-110.9467	-4.6564	-37.2%	-20.8%
$p=12$	-120.1037	-13.8134	-74.9%	-49.9%

- $\hat{\Sigma}$ is the covariance matrix of the RF-VAR residuals
- Evaluated on the common sample of 159 observations

Standard Deviation Ratios ReducedForm

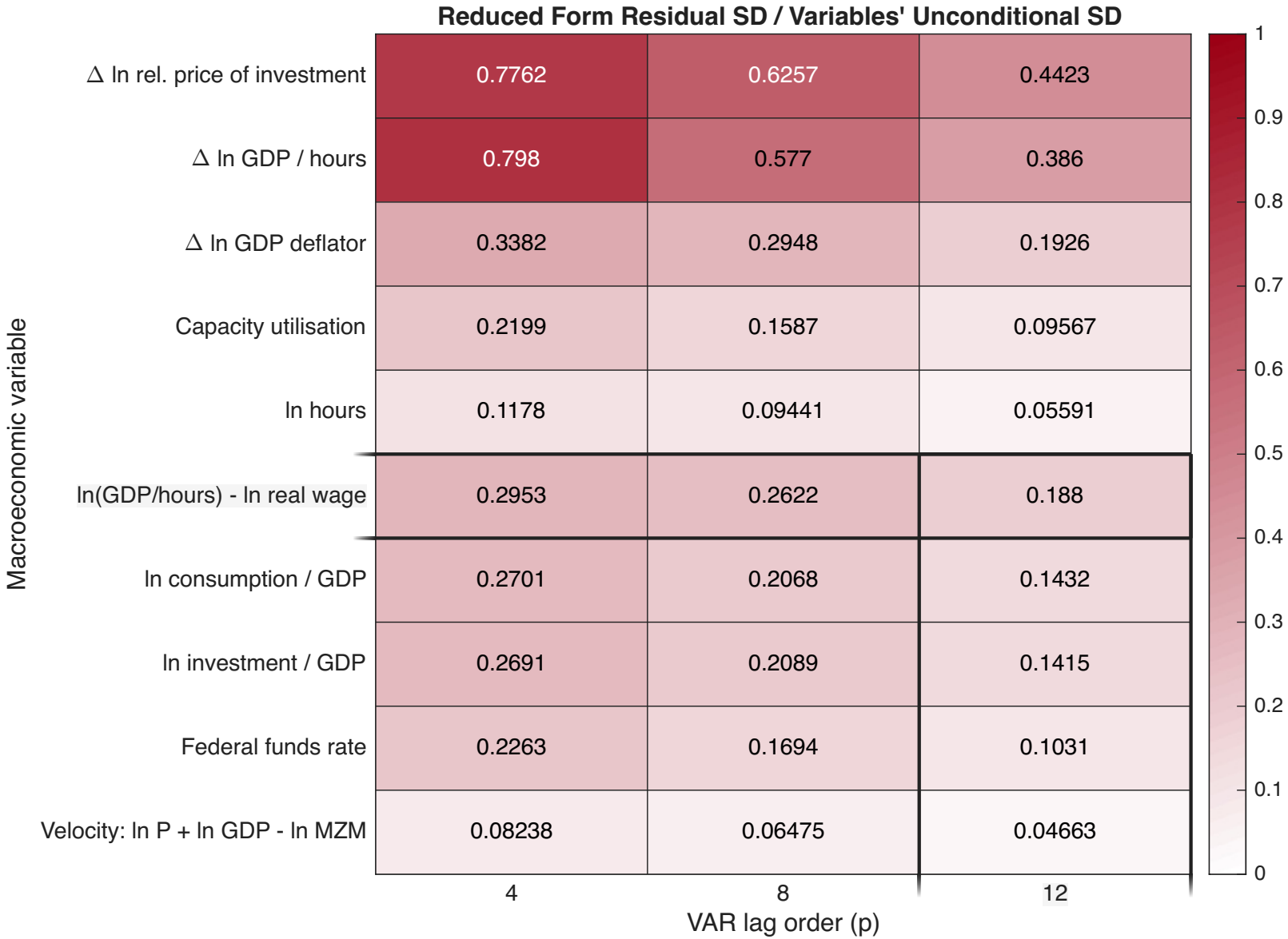


Figure 9: Standard Deviation Ratios ReducedForm

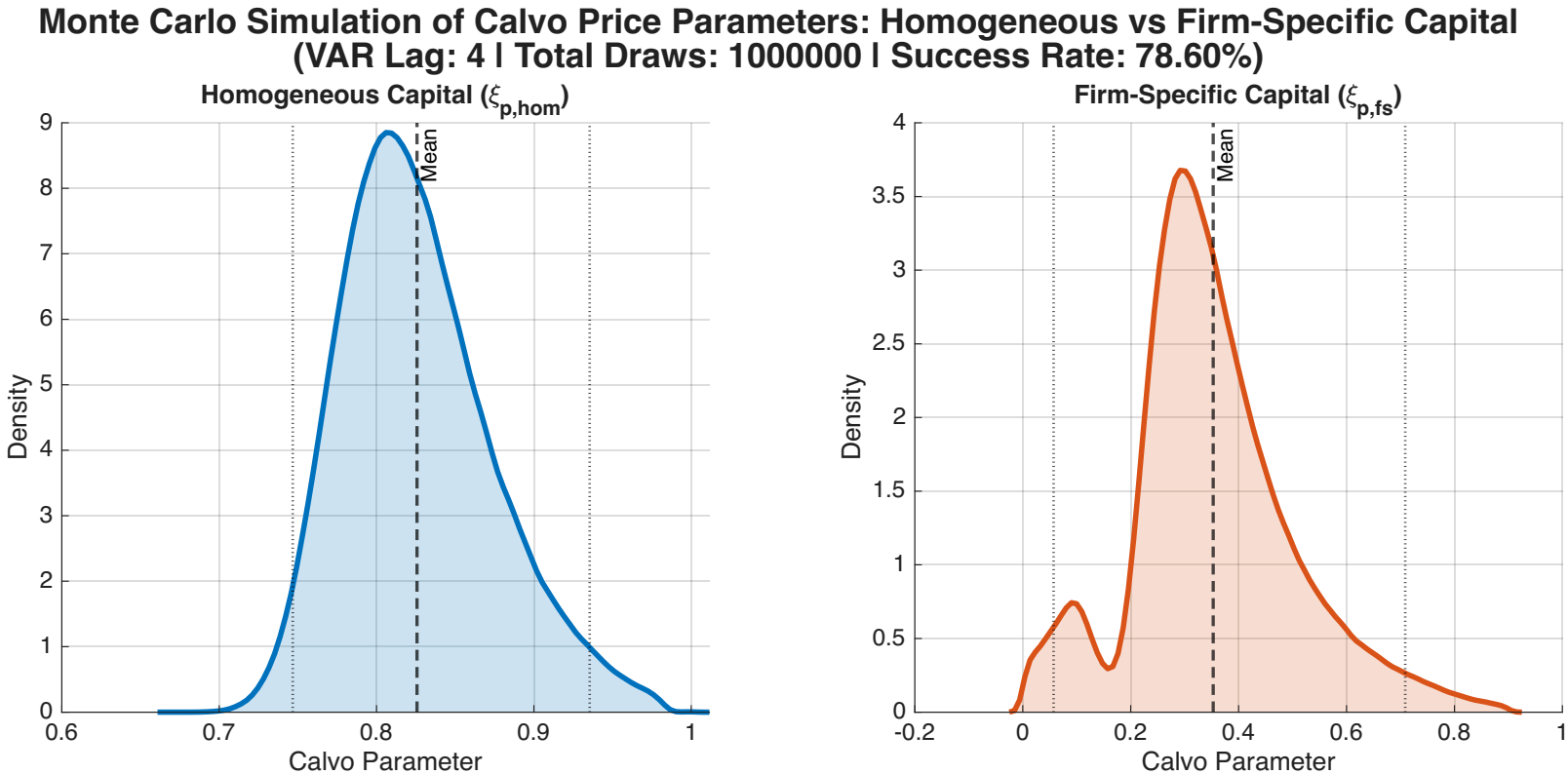


Figure 10: Calvo Price Parameter PDF from Montecarlo Simulation; VAR(4)

MonteCalvo

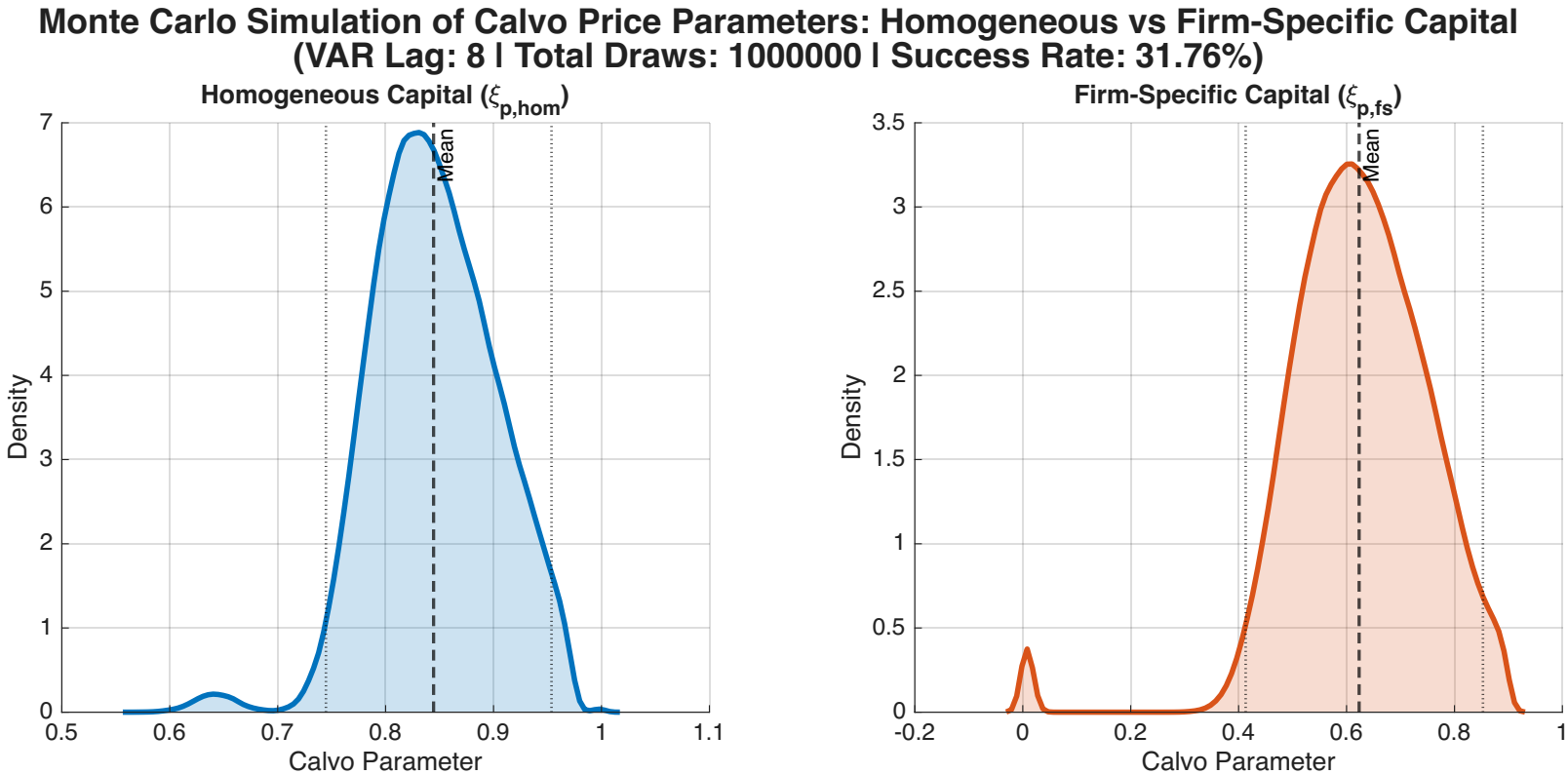


Figure 11: Calvo Price Parameter PDF from Montecarlo Simulation; VAR(8)

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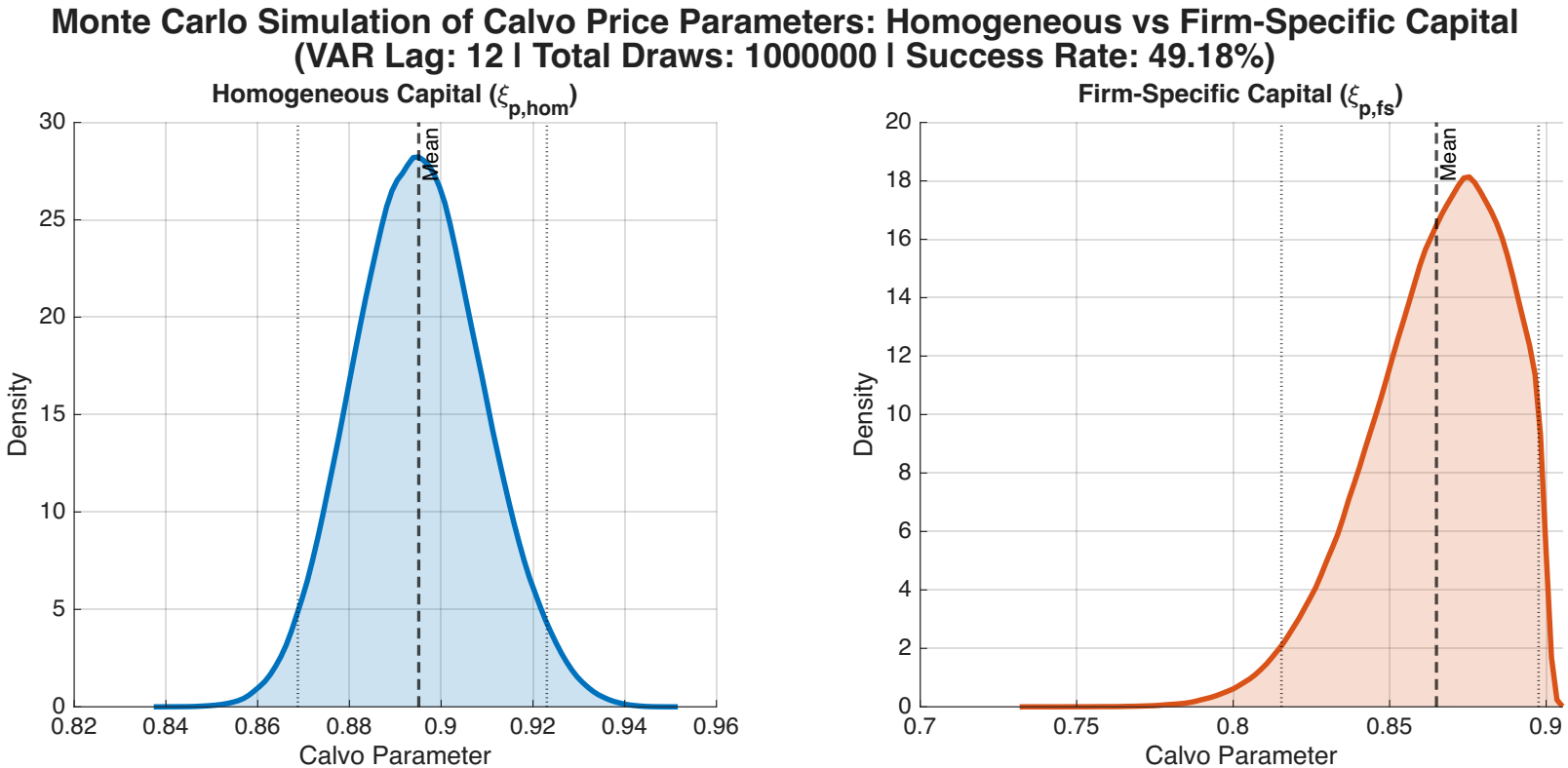


Figure 12: Calvo Price Parameter PDF from Montecarlo Simulation; VAR(12)